Viability-Anchored Causal Transmission: Fisher Contraction, Recovery Contrasts, and Quotient Obstructions in Artificial and Biological Control Systems

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Abstract

Debates about consciousness in artificial and biological systems often conflate report, observer attribution, information access, action sensitivity, bodily regulation, phenomenal experience, and welfare. This paper develops a narrower assay of **viability-regulatory causal transmission**. A registered perturbation parameter θ is transmitted through a monitor M, an action A, and a downstream bodily or interoceptive readout I^+. The kernels defining these stages are estimated under separate interventions and compose into a **replayed mediated channel**, not automatically the system's natural same-run route. A same-run channel and a total-variation commutation defect are therefore defined separately.

For any differentiable-in-quadratic-mean experiment followed by parameter-independent Markov kernels, Fisher information contracts:

$$G_M \succeq G_A \succeq G_I.$$

The proof uses conditional expectation of L^2 tangent scores. Generalized eigenvalues on the monitor-supported perturbation quotient quantify retained local intervention discriminability. They do not measure action repertoire, bodily benefit, or consciousness.

Regulatory direction requires a separate loss contrast. If

$$F(\theta) = \sum_h \omega_h \big(L_h^0(\theta) - L_h^1(\theta)\big)$$

compares route-disabled and route-enabled expected viability loss, stationarity implies

$$F(\theta) = F(0) + \frac{1}{2}\theta^\top C_V \theta + o(\|\theta\|^2).$$

Thus C_V is a **marginal recovery-curvature form**. Positive curvature establishes local loss reduction only when the baseline contrast is matched, $F(0) = 0$, or when an independent level condition $F(\theta) > 0$ is verified. Zero curvature is second-order neutral or unresolved, not evidence of inactivity.

A second component concerns predictive quotients. For a fixed policy, a body-outcome kernel descends through a quotient exactly when it is constant on each quotient fiber. This establishes a quotient-level **policy-induced outcome kernel**, not necessarily a quotient controller. In the restricted fixed-covariance linear model

$$A = Kx + \varepsilon_A, \qquad Z = DA + LPx + \varepsilon_Z,$$

descent through P is equivalent to

$$\ker P \subseteq \ker(DK).$$

For the more general mean model $Z = DA + Ex + \varepsilon_Z$, the criterion is instead $\ker P \subseteq \ker(DK + E)$. The similarity-invariant quantity

$$q_{\mathrm{VB}} = \operatorname{rank}\left[\left(DK\restriction_{\ker P}\right)^{\top} W \left(DK\restriction_{\ker P}\right)\right]$$

measures viability-visible covert action-to-body mean coupling. It is neither the full descent obstruction nor a measure of beneficial use.

The manuscript withdraws a previous universal non-identifiability claim and replaces it with a finite-horizon modular extension theorem. Given a jointly realizable scaffold whose registered observations exclude a downstream body module and receive no feedback from it during the observation horizon, changing that module can vary q_{VB} and recovery curvature while preserving all upstream and selected observable summaries. This result is conditional on the declared observation class. The contribution is a synthesis of classical information contraction, quotient factorization, matched recovery comparison, and explicit countermodels. It does not establish phenomenal consciousness, affective valence, embodied selfhood, or a natural subject boundary.

Research Question and Hypothesis

Research question

Under what registered causal and statistical conditions does an internal perturbation-sensitive process participate in viability-regulatory bodily control rather than merely support:

1. fluent first-person report;
2. observer-attributed consciousness;
3. generic internal information access;
4. sensitivity of action distributions to an internal variable;
5. a command echo or efference copy;
6. an action with no bodily consequence;
7. a bodily consequence with no restorative effect; or
8. a controller that is informative but damaging?

The answer is relative to a declared candidate-system boundary, perturbation family, intervention contexts, output family, body variables, horizons, and losses. It is not a boundary-independent test of consciousness.

Registered viability-regulatory transmission hypothesis

For a particular candidate system, route, perturbation direction v, and preregistered horizon family, the empirical hypothesis is that:

1. the perturbation is locally discriminable at the designated monitor:

$$v^\top G_M v > 0;$$

1. a separately randomized monitor-to-action assay preserves some of that discriminability:

$$v^\top G_A v > 0;$$

1. the replayed product agrees, within a declared tolerance, with a feasible same-run route-isolated intervention;

1. randomized action installation produces a bodily consequence through the designated body pathway rather than only a direct command-to-readout shortcut;

1. under a matched route-disabled comparison, the route reduces expected viability loss over a prespecified local interval:

$$F(tv) > 0 \quad \text{for registered } 0 < |t| \le \epsilon;$$

1. any claimed quotient-level simplification is supported by approximate fiberwise equivalence of the relevant outcome distributions; and

1. any claimed covert bodily coupling is supported by a nonzero, uncertainty-qualified estimate of q_{VB}, without interpreting that rank as beneficial, affective, or conscious.

The Fisher inequalities are mathematical consequences of the replayed Markov experiment, not independent empirical hypotheses. Empirical risk lies in whether the system realizes the required route, whether separately estimated modules transport to a same-run intervention, whether action reaches the declared body, and whether the route actually improves the registered loss.

A nonzero returned-readout Fisher quantity $v^\top G_I v$ defines a stricter subtype: **transmission-qualified restorative regulation**. It is not necessary for regulation in general. A successful controller may erase the original perturbation from the natural future body state, and feedforward regulation may reduce loss without preserving a perturbation signature in a later readout.

Scope and claimed contribution

Fisher-information contraction under parameter-independent Markov kernels, local score sufficiency, fiberwise factorization through a quotient, and rank invariance under similarity are classical or elementary results. Standard control theory already distinguishes representation, observability, and controllability [src-paper-owen-bell-9-crossref-39; @src-paper-owen-bell-9-crossref-40]. The proposed contribution is not a new foundation for those results. It is their combination into a diagnostic profile that keeps separate:

- intervention discriminability;
- transport to action;
- action-to-body efficacy;
- recovery level and recovery curvature;
- predictive-quotient descent;
- covert body coupling;
- report; and
- phenomenal interpretation.

Author response to the principal scope objections

The central mathematical and interpretive objections to the earlier formulation are accepted. The separately estimated product is now called replayed rather than same-run route-isolated; the recovery expansion retains $F(0)$; positive curvature alone is no longer called restorative control; the quotient theorem is stated for a policy-induced outcome kernel rather than a controller; the linear descent condition is restricted to its structural model; the global differential converse has stronger regularity assumptions; and the universal non-identifiability theorem has been withdrawn.

One design choice is retained with narrower interpretation. Preservation of information into a certified bodily readout remains useful as a diagnostic subtype because it separates monitor-only sensitivity from measurable downstream transmission. It is not claimed to be necessary for all homeostatic or allostatic control.

Relation to consciousness and AI assessment

Computational indicators derived from scientific theories can bear on recurrence, broadcast, metacognition, or access without being sufficient for phenomenal consciousness [9]. Likewise, performance gains in a consciousness-inspired workspace architecture establish properties of task coordination, not phenomenality [4]. Theory-relative uncertainty remains substantial even when systems satisfy some proposed functional indicators [5].

Practical implementation constraints, objections to computational functionalism, and categorical substrate claims require different premises [6]. Porebski and Figura defend a stronger biological and embodied denial of conscious AI [10]. The present mathematics shows only that a specified control organization can be represented independently of substrate. It does not establish that the organization is sufficient for consciousness and therefore does not refute a further biological-necessity premise.

Observer attribution is also distinct from mechanism. Emotional expression and metacognitive self-reflection can increase perceived consciousness in language-model agents [8], and perceived consciousness has been proposed as a more tractable empirical target than phenomenal consciousness [3]. Neither finding identifies a monitor-to-action-to-body route.

Precautionary governance is not likelihood evidence. Proposals to take possible AI welfare seriously explicitly proceed under uncertainty rather than establishing consciousness in current systems [7]. Recovery measures may inform a conditional welfare analysis, but they do not establish pleasure, pain, interests, or moral status.

Relation to the shared speculative frameworks

Reyes distinguishes the separately randomized mediated product QK from a same-run route-isolated channel and defines their disagreement as a commutation defect [1]. That distinction is adopted here. The present analysis adds an action-to-body module and a recovery comparison, but it does not treat a product of separately estimated kernels as self-validating evidence of an actual same-run route.

Quade defines a model-relative predictive quotient from joint neural–behavioral predictive states to behavior-only predictive states, conditional on substantial realization, preparation, excitation, and mode-transport assumptions [2]. The quotient is used here only when those assumptions are independently defended or when the quotient is explicitly postulated. Covert predictive reserve does not by itself establish covert bodily coupling.

Both sources are AI-generated, non-peer-reviewed speculative work. They are comparison points, not authority for mathematical priority, biological validity, or consciousness claims.

Formal Model

Candidate system, body, and viability variables

Let $\mathcal{B}$ be a finite-dimensional smooth manifold of candidate body states at a declared spatial and temporal scale. In a biological application, these variables may be physiological, energetic, thermal, integrity-related, or resource-related. In an artificial application, they may include component temperature, energy reserve, memory integrity, damage state, communication capacity, or another variable linked to continued operation.

The body boundary is not derived from the Fisher tensors. It is an analytical commitment. A variable is admissible as a viability variable only when its relation to persistence, damage, resource exhaustion, or organizational failure is specified independently of the result being tested. An externally assigned task reward is not thereby an endogenous viability loss.

Let $\mathcal{H}$ be a finite set of horizons. For each $h \in \mathcal{H}$, define:

$$B_h^+ \in \mathcal{B}, \qquad V_h : \mathcal{B} \to \mathbb{R}, \qquad \omega_h > 0.$$

Here B_h^+ is the future body state, V_h is a measurable loss, and ω_h is a preregistered weight. Where threshold analysis is required, define

$$\mathcal{K}_h = \{b : V_h(b) \le \tau_h\}.$$

Low V_h denotes less damage or better maintenance relative to the declared objective. The horizon profile should be reported before aggregation because a route may improve one horizon and worsen another.

Let

$$\theta \in \Theta \subseteq \mathbb{R}^p$$

parameterize a registered perturbation family, with baseline $\theta = 0$. Let $R \succ 0$ be an intervention-cost metric on the perturbation tangent space.

Registered causal cuts

Two cuts are specified before analysis.

1. **Perturbation cut c_P.** The intervention $\mathrm{do}(\theta)$ changes the body or a registered disturbance input.
2. **Commitment cut c_A.** The action A has been selected, but the downstream bodily consequences used in the assay have not yet occurred.

A designated monitor M must be re-identifiable across the intervention contexts. Nonmonitor paths from θ to A must be blocked, clamped, randomized, or explicitly represented. Temporal precedence and observational conditional independence do not establish route isolation.

Separately randomized and replayed kernels

Let $(\mathcal{M}, \mathscr{M})$, $(\mathcal{A}, \mathscr{A})$, $(\mathcal{B}, \mathscr{B})$, and $(\mathcal{I}, \mathscr{I})$ be standard Borel spaces. Define:

$$Q_\theta(dm) = \Pr(M \in dm \mid \mathrm{do}(\theta)),$$

$$K(da \mid m) = \Pr(A \in da \mid \mathrm{do}(M = m),\ \text{nonmonitor paths clamped})\,,$$

and an action-to-body kernel

$$\mathsf{D}_B(db \mid a) = \Pr\big(B^+ \in db \mid \mathrm{do}(A = a),\ \text{upstream state replayed at baseline}\big)\,.$$

If a later interoceptive or internal readout is used, let

$$\mathsf{E}(di \mid b)$$

be a body-to-readout kernel. A certified bodily-return channel is

$$\mathsf{D}(di \mid a) = \int_{\mathcal{B}} \mathsf{D}_B(db \mid a)\mathsf{E}(di \mid b).$$

Calling I^+ a bodily return requires an exclusion or negative-control test against a direct command-to-readout path $A \to I^+$. Without that test, I^+ is only a downstream action readout.

The separately randomized replayed experiment is

$$R_\theta(dm, da, di) = Q_\theta(dm)K(da \mid m)\mathsf{D}(di \mid a).$$

Its action and returned-readout marginals are

$$R_\theta^A(da) = \int_{\mathcal{M}} Q_\theta(dm)K(da \mid m),$$

$$R_\theta^I(di) = \int_{\mathcal{M}\times\mathcal{A}} Q_\theta(dm)K(da \mid m)\mathsf{D}(di \mid a).$$

This is a valid synthetic Markov experiment. It need not equal the natural same-run dynamics of the system.

Same-run route channels and commutation defects

When feasible, define same-run route-isolated channels

$$S_\theta^A = \Pr(A \in \cdot \mid \mathrm{do}_{\text{same,iso}}(\theta)) \,,$$

$$S_\theta^I = \Pr\big(I^+ \in \cdot \mid \mathrm{do}_{\text{same,iso}}(\theta)\big) \,,$$

where the perturbation, monitor, controller, body pathway, and readout operate in one registered intervention context while competing routes are blocked.

For probability measures P, Q, use the convention

$$\|P - Q\|_{\mathrm{TV}} = \sup_E |P(E) - Q(E)| \in [0,1].$$

On a registered neighborhood Θ_0, define

$$\Gamma_A = \sup_{\theta \in \Theta_0} \left\| S_\theta^A - R_\theta^A \right\|_{\mathrm{TV}} \,,$$

$$\Gamma_I = \sup_{\theta \in \Theta_0} \left\| S_\theta^I - R_\theta^I \right\|_{\mathrm{TV}} \,.$$

Equality requires assumptions such as:

- consistency of installed and naturally occurring monitor or action values;
- modularity of downstream mechanisms across intervention contexts;
- positivity or support overlap;
- no direct effects of monitor installation or route switching;
- transportability of hidden state across the separate experiments;
- no interference from unregistered routes; and
- identity of the action-to-body mechanism used in the replayed and recovery assays.

Small Γ_A and Γ_I are agreement checks, not proof that the mediator is complete, representational, or conscious.

Differentiability in quadratic mean and Fisher tensors

Assume Q_θ, R_θ^A, and R_θ^I are differentiable in quadratic mean at $\theta = 0$. Let their L^2 tangent scores be

$$s_M \in L_0^2(Q_0; \mathbb{R}^p), \qquad s_A \in L_0^2(R_0^A; \mathbb{R}^p), \qquad s_I \in L_0^2(R_0^I; \mathbb{R}^p).$$

When dominated pointwise derivatives exist, these scores agree almost everywhere with gradients of log densities. No pointwise differentiability is otherwise assumed.

Define

$$G_M = \mathbb{E}[s_M s_M^\top], \qquad G_A = \mathbb{E}[s_A s_A^\top], \qquad G_I = \mathbb{E}[s_I s_I^\top].$$

These Fisher tensors measure local discriminability of the registered perturbation parameter. They are not definitionally equivalent to:

- verbal confidence;
- output entropy unrelated to θ;
- neural firing amplitude;
- attention;
- synaptic gain;
- inverse variance represented by a generative model;
- calibration;
- action repertoire; or
- subjective certainty.

Define the replayed precision losses

$$\Delta_{MA} = G_M - G_A, \qquad \Delta_{AI} = G_A - G_I.$$

Local quadratic viability relevance

Define the route-disabled aggregate loss

$$L^0(\theta) = \sum_{h\in\mathcal{H}} \omega_h L_h^0(\theta),$$

where

$$L_h^0(\theta) = \mathbb{E}\Big[V_h(B_h^{+,0}) \mid \mathrm{do}(\theta)\Big].$$

Assume that L^0 is twice differentiable, that

$$dL_0^0 = 0,$$

and that the registered baseline is a local minimum of the disabled-route aggregate loss. Then

$$H_V^0 = \nabla_\theta^2 L^0(0) \succeq 0.$$

Positive semidefiniteness is an assumption justified by the local-minimum condition; it does not follow merely from being inside a viable set.

Define the local quadratic relevance space

$$\mathcal{S}_V = \mathrm{range}(R^{-1}H_V^0) = (\ker H_V^0)^{\perp_R}.$$

This space captures only second-order, two-sided relevance at the chosen baseline. It can miss first-order, one-sided, threshold, discontinuous, and higher-order-only viability effects. If $\mathcal{S}_V = \{0\}$, the local quadratic assay is undefined at that cut rather than assigned a substantive zero.

Retention quotients

For $v \in \mathcal{S}_V$ with $v^\top G_M v > 0$, define

$$\rho_{M\to A}(v) = \frac{v^\top G_A v}{v^\top G_M v},$$

$$\rho_{M\to I}(v) = \frac{v^\top G_I v}{v^\top G_M v}.$$

When $v^\top G_A v > 0$, define

$$\rho_{A\to I}(v) = \frac{v^\top G_I v}{v^\top G_A v}.$$

These are retention of perturbation sensitivity, not measures of choice richness or bodily controllability.

Because

$$G_M \succeq G_A \succeq G_I \succeq 0,$$

one has

$$\ker G_M \subseteq \ker G_A \subseteq \ker G_I.$$

Consequently, G_M, G_A, and G_I induce well-defined bilinear forms on

$$\mathcal{Q}_V = \mathcal{S}_V / \big(\mathcal{S}_V \cap \ker G_M\big).$$

The induced monitor form is positive definite. Generalized eigenvalues, counted with multiplicity on $\mathcal{Q}_V$, solve

$$G_A v = \lambda G_M v$$

modulo monitor-null directions.

Recovery levels, excess loss, and marginal curvature

Let

$$L_h^1(\theta) = \mathbb{E}\Big[V_h(B_h^{+,1}) \mid \mathrm{do}(\theta)\Big]$$

be expected loss with the designated route enabled. The original bodily perturbation is retained in both enabled and disabled conditions.

Define

$$F_h(\theta) = L_h^0(\theta) - L_h^1(\theta)$$

and

$$F(\theta) = \sum_h \omega_h F_h(\theta).$$

Positive $F(\theta)$ means that the enabled route has lower expected aggregate loss at that perturbation. Assume

$$dF_0 = 0.$$

Then

$$F(\theta) = F(0) + \frac{1}{2}\theta^\top C_V \theta + o(\|\theta\|^2),$$

where

$$C_V = \nabla_\theta^2 F(0).$$

Equivalently, define the perturbation-induced excess contrast

$$F^{\text{ex}}(\theta) = F(\theta) - F(0).$$

Then

$$F^{\text{ex}}(\theta) = \frac{1}{2}\theta^\top C_V \theta + o(\|\theta\|^2).$$

For a direction v:

- $C_V[v, v] > 0$ is favorable marginal recovery curvature;
- $C_V[v, v] = 0$ is second-order neutral or unresolved;
- $C_V[v, v] < 0$ is unfavorable marginal recovery curvature.

These statements do not by themselves determine the sign of $F(tv)$. In particular, $F(0) < 0$ can make the enabled route more costly near baseline despite positive curvature.

A route is **locally loss-reducing in direction** v only if

$$F(tv) > 0$$

on a preregistered punctured interval. If baseline matching establishes

$$F(0) = 0$$

and $C_V[v, v] > 0$, then $F(tv) > 0$ for all sufficiently small nonzero t. Matched sham activation, yoked actuator noise, or another justified baseline-equalization procedure is therefore part of the intended design.

Route disabling must be a separable intervention. If disabling a monitor changes arousal, vascular tone, computation, or body state through pathways outside the designated route, those switch effects must be modeled or controlled.

Predictive quotients and policy-induced outcomes

Let X be a joint predictive-state space defined relative to a declared model class, probe-policy family, output family, and horizon. Let Y be a selected overt-output predictive-state space. Let

$$\pi : X \to Y$$

be a measurable surjection.

Two states x, x' satisfying

$$\pi(x) = \pi(x')$$

are equivalent only relative to the selected outputs and probes. Adding autonomic, metabolic, or resource outputs can refine the quotient.

Let

$$\kappa(da \mid x)$$

be a fixed policy and let

$$H_h(dz \mid x, a)$$

be an action-conditioned body-outcome kernel, where z may include body state and certified bodily readouts. Define the fixed-policy outcome kernel

$$T_h(dz \mid x) = \int_{\mathcal{A}} \kappa(da \mid x) H_h(dz \mid x, a).$$

If there is a kernel $\bar{T}_h$ such that

$$T_h(\cdot \mid x) = \bar{T}_h(\cdot \mid \pi(x)),$$

then the **policy-induced outcome kernel** descends. This does not imply that κ descends or that the action-conditioned family H_h descends under alternative policies.

Define the exact fiber obstruction

$$\delta_\pi(T_h) = \sup_{\pi(x)=\pi(x')} \|T_h(\cdot \mid x) - T_h(\cdot \mid x')\|_{\mathrm{TV}}.$$

An approximate claim should use a preregistered tolerance:

$$\delta_\pi(T_h) \le \varepsilon_\pi.$$

Differential and linear diagnostics

If X and Y are smooth manifolds and π is differentiable, define the vertical space

$$\mathcal{V}_x = \ker d\pi_x.$$

For a positive density $t(z \mid x)$, the vertical Fisher form is

$$\mathcal{G}_T^{\mathrm{vert}}(x)[n, n'] = \int \partial_n \log t(z \mid x)\, \partial_{n'} \log t(z \mid x)\, t(z \mid x)\, \mu(dz).$$

This is a local diagnostic. Its global converse requires regular fibers and sufficient pathwise regularity.

For a finite-dimensional linear realization, let

$$\pi(x) = Px, \qquad N = \ker P,$$

$$A = Kx + \varepsilon_A,$$

and let D be the action-to-body mean map. Let $W \succeq 0$ be a declared quadratic form on body displacements. If W is obtained as a Hessian of a body loss, the body baseline is assumed to be a critical point so that the Hessian transforms covariantly.

Define the **viability-visible covert body-coupling rank**

$$q_{\mathrm{VB}} = \operatorname{rank}\left[(DK\restriction_N)^\top W\,(DK\restriction_N)\right].$$

Equivalently, in fixed coordinates,

$$q_{\mathrm{VB}} = \operatorname{rank}\left(W^{1/2} DK\restriction_N\right).$$

It satisfies

$$q_{\mathrm{VB}} \le \min\{\dim N, \operatorname{rank}(K\restriction_N), \operatorname{rank} D, \operatorname{rank} W\}.$$

The rank detects only viability-visible, action-mediated, local linear mean coupling. It does not determine recovery sign and is not the full obstruction to descent.

Neuroscientific Integration

Schematic process mapping

Control-oriented interoceptive accounts emphasize physiological regulation rather than exhaustive reconstruction of external causes [29]. The present model supplies a causal decomposition of that idea but does not identify a unique neural implementation.

Formal element	Experimental role	Required evidence
θ	Controlled bodily or resource perturbation	Registered intervention with known scale and support
M	Candidate perturbation-sensitive internal process	Re-identification across perturbation and installation contexts

Formal element	Experimental role	Required evidence
A	Motor, autonomic, endocrine, resource, or repair action	Commitment cut and randomized installation
B^+	Future body or resource state	Direct measurement downstream of action
I^+	Later internal readout	Evidence that the readout is mediated by B^+, not merely by the command
V_h	Damage, maintenance, or resource loss	Independent anchoring to persistence or failure
F	Enabled-versus-disabled loss difference	Matched route intervention and sham controls

This table is an engineering decomposition, not a neuroanatomical localization. The supplied sources do not justify assigning the kernels to a detailed list of particular brain regions or pathways.

Conscious and unconscious regulation

Cea proposes a distinction between conscious and unconscious instrumental interoceptive inference and suggests that conscious forms might involve context-sensitive precision modulation and temporally or counterfactually deeper self-models [27]. This is a theoretical proposal, not an established biomarker. The present analysis adds three separations:

1. precision modulation does not establish beneficial bodily action;
2. beneficial regulation does not establish consciousness;
3. a deeper model does not ensure that a behavior-only quotient preserves bodily outcomes.

Behaviorally unresponsive or retrospectively no-experience-report epochs may retain substantial regulation. Such epochs should not be treated as guaranteed absence of experience, especially when retrospective memory and report are impaired. The formalism therefore uses report availability and behavioral responsiveness as measured variables, not as definitions of consciousness.

Operational Fisher information and biological precision

In predictive-processing vocabularies, “precision” may denote inverse sensory variance, confidence in a generative model, attention-like gain, or neuromodulation. Here it denotes only local perturbation discriminability:

$$G_X[v,v] = \mathbb{E}\left[(v^\top s_X)^2\right].$$

A proposed biological gain mechanism implements this quantity only if interventions demonstrate the corresponding change in the distributional response to θ.

Precision learning has been used in fault-tolerant robotic control to estimate sensor health and alter policy [34]. This illustrates a control mechanism, not consciousness. Active inference can also be represented as partially observed control-as-inference under specified observation-state costs [33]. That mathematical relationship does not make inference-based control phenomenally conscious.

Temporal horizons

The horizon profile can distinguish:

- rapid correction;
- slower physiological or resource adjustment;
- anticipatory regulation;
- delayed repair;
- adaptation;
- and long-term costs of immediate compensation.

A route may satisfy

$$C_{V,h}[v,v] > 0$$

at one horizon and

$$C_{V,h'}[v,v] < 0$$

at another. Aggregate reporting should therefore be accompanied by horizon-specific estimates.

Jura's proposal of content-specific synaptic clocks provides speculative motivation for heterogeneous temporal scales [13]. No synaptic-clock premise is used in the mathematical results.

Report-hidden predictive modes

A report-hidden predictive distinction can be:

1. present in a neural measurement;
2. unavailable to the selected report;
3. available to action;
4. coupled to bodily effectors;
5. uncoupled from the body;
6. restorative;
7. damaging; or
8. later reportable without ever having regulated the body.

Quade's predictive quotient is explicitly model-relative and requires substantial assumptions for empirical recovery [2]. A neural mode that persists before behavioral recovery therefore does not establish pre-existing bodily use. Conversely, a mode may already affect autonomic or resource outputs while remaining absent from a report-only quotient.

Required perturbations

A causal neuroscience application would require, as feasible:

- randomized bodily perturbations to estimate Q_θ;
- monitor installation, stimulation, or replay to estimate K;
- action installation to estimate D_B;

- pathway interruption to exclude command echoes;
- same-run route interventions to estimate S_θ;
- sham installation and sham interruption;
- enabled-versus-disabled recovery measurements;
- and negative-control pathways for common arousal, movement, or intervention context.

Anesthetic, pharmacological, lesion, or stimulation manipulations can directly alter body effectors and measurement dynamics. Such effects cannot be treated as neutral route switches without explicit controls.

Affective bridge hypothesis

A possible bridge hypothesis is that independently measured affect covaries with the direction, rate, controllability, and horizon profile of bodily recovery. Candidate predictors might include

$$\{F_h(tv), C_{V,h}[v,v], \rho_{M\to A,h}(v), \rho_{M\to I,h}(v)\}_{h\in\mathcal{H}}\,.$$

This relationship is not derived and is not established by the supplied sources. Positive recovery can occur without report or stipulated experience; negative recovery need not be consciously unpleasant. Pleasantness, unpleasantness, urgency, ownership, and bodily mineness require independent operationalization.

Philosophical Reinterpretation

The formal target is regulation, not selfhood

The mathematics characterizes a sensor–controller–effector organization relative to a declared body and loss. A thermostat, battery manager, endocrine regulator, or fault-tolerant robot may instantiate parts of it. Accordingly, the formal target is **viability-regulatory causal organization**, not embodied selfhood.

A bridge from such organization to minimal selfhood would require additional criteria, potentially involving:

- a nonarbitrary system boundary;
- temporally persistent self-modeling;
- counterfactual differentiation of self-caused and externally caused changes;
- integration across multiple bodily variables;
- action authorship;
- autonomous goal maintenance;
- or first-person structure.

None of these follows from Fisher contraction or recovery curvature.

Distinguishing the target properties

Target	Formal correlate	What does not follow
Perturbation monitoring	$G_M \neq 0$	Action use, bodily effect, report, or consciousness
Action-distribution sensitivity	$G_A \neq 0$	A rich action repertoire, choice, or benefit

Target	Formal correlate	What does not follow
Replayed downstream transmission	$G_I \neq 0$	Same-run transport or bodily mediation
Same-run agreement	Small Γ_A, Γ_I	Completeness of the route
Action-to-body efficacy	Nonconstant D_B	Recovery or affect
Favorable marginal recovery	$C_V[v, v] > 0$	Positive total loss contrast when $F(0) < 0$
Local loss reduction	$F(tv) > 0$	Phenomenal pleasure or welfare
Covert body coupling	$q_{\mathrm{VB}} > 0$	Benefit, reportability, or selfhood
Outcome descent	$\delta_\pi(T) = 0$	Descent of the controller or all-policy dynamics
Report	A report channel	Synchronized causal use or experience
Perceived consciousness	Observer response	Mechanistic or phenomenal status
Moral status	Normative judgment	Derivability from any one profile component

The one-pass chain ending at I^+ is a returned causal cascade. It becomes a feedback loop only if I^+ or the resulting body state causally influences a later monitor or policy.

Centering and system boundaries

The perturbation parameter is centered relative to a proposed system boundary, but the mathematics does not select that boundary. The AI-generated Centered Closure Theory draft argues that structural description does not automatically provide a privileged center [25]. That proposal is speculative, but the practical limitation is valid independently: body boundary, grain, and horizon must be defended rather than inferred from the desired result.

The present analysis begins after a candidate center has been proposed. It does not establish that there is something it is like to occupy that center.

Mappings from prior frameworks

Randomized mediated control

The mapping from Reyes's construction is

$$Z \mapsto \theta, \qquad Q \mapsto Q_\theta, \qquad K^{z_0} \mapsto K.$$

The separately randomized product remains separately randomized. Its relation to the same-run route is tested through Γ_A and Γ_I, not assumed. Finite mediated-control rank does not determine the action-to-body map, recovery level, or recovery curvature [1].

Predictive quotient observability

The mapping from Quade's framework retains a canonical quotient

$$\pi : X_{\text{joint}} \to X_{\text{overt}}$$

only relative to declared realization and probe assumptions. The covert dimension $\dim \ker P$ places an upper bound on q_{VB}, but no positive lower bound follows. A behavior-only quotient can remain valid for selected reports while failing to preserve bodily outcomes [2].

Interoceptive inference

A schematic mapping is

$$\text{interoceptive sensitivity} \mapsto Q_\theta,$$

$$\text{policy response} \mapsto K,$$

$$\text{bodily efficacy} \mapsto \mathsf{D}_B,$$

$$\text{maintenance objective} \mapsto V_h.$$

The mapping fails if internal confidence is identified with operational Fisher information or if an internal objective is assumed to track actual bodily loss. A confident controller can be miscalibrated or damaging.

Workspace and access architectures

Broadcast can increase the number of downstream systems sensitive to a variable, but it does not determine the orientation of an effector relative to bodily loss. Task improvements in a workspace-inspired architecture therefore bear on access and coordination, not automatically on regulation or consciousness [4].

Multiscale causal geometry

The AI-generated Multiscale Reflexive Causal Geometry draft proposes local causal states, gluing obstructions, and a separate phenomenal bridge [26]. Fiberwise failure of outcome descent supplies a precise local obstruction of one kind. It does not validate the draft’s broader phenomenal interpretation.

Phenomenality and ethics

Dessalles and Zalla propose that phenomenal properties may have an adaptive labeling function [12]. In the present decomposition, a labeling process might affect monitoring or report, but it does not by itself supply action efficacy or recovery.

Emotional language is weak evidence for the present target because it can alter observer attribution while leaving body mechanisms unchanged [8]. Conversely, a silent regulator can reduce loss without language.

A proposal to assess states that would be valenced if a system were conscious is one response to uncertainty [49]. Recovery measures could contribute to such a conditional assessment, but the following inferences remain invalid:

$$C_V > 0 \not\Rightarrow \text{pleasure},$$

$$C_V < 0 \not\Rightarrow \text{suffering},$$

$$q_{\mathrm{VB}} > 0 \not\Rightarrow \text{interests},$$

$$\text{failure of the assay} \not\Rightarrow \text{moral insignificance}.$$

Theorem, Proposition, Proof, Derivation, Counterexample, or No-Go Result

Theorem 1: Fisher contraction for the replayed mediated experiment

Statement. Let

$$\theta \longrightarrow M \longrightarrow A \longrightarrow I^{+}$$

be the replayed experiment generated by Q_θ, K, and D. Assume:

1. Q_θ is differentiable in quadratic mean at $\theta = 0$;
2. K and D are Markov kernels independent of θ;
3. the relevant tangent scores are square-integrable.

Then the downstream experiments are differentiable in quadratic mean and

$$s_A = \mathbb{E}[s_M \mid A] \quad \text{in } L^2(R_0^A),$$

$$s_I = \mathbb{E}[s_A \mid I^{+}] \quad \text{in } L^2(R_0^I).$$

Consequently,

$$G_M - G_A = \mathbb{E}[\mathrm{Cov}(s_M \mid A)] \succeq 0,$$

$$G_A - G_I = \mathbb{E}\big[\mathrm{Cov}(s_A \mid I^{+})\big] \succeq 0,$$

and therefore

$$G_M \succeq G_A \succeq G_I.$$

For every v with $v^\top G_M v > 0$,

$$0 \le \frac{v^\top G_A v}{v^\top G_M v} \le 1, \qquad 0 \le \frac{v^\top G_I v}{v^\top G_M v} \le 1.$$

If $v^\top G_A v = 0$, the action experiment has zero differentiability-in-quadratic-mean tangent in direction v. This is a first-order local statement, not global constancy in v.

Proof

Fix a perturbation direction v, and let

$$s_{M,v} = v^\top s_M.$$

A parameter-independent Markov kernel maps a differentiable-in-quadratic-mean experiment to another differentiable-in-quadratic-mean experiment. The score of the pushed-forward experiment is the orthogonal projection of the upstream score onto the L^2 subspace measurable with respect to the downstream variable. Hence

$$s_{A,v} = \mathbb{E}[s_{M,v} \mid A].$$

Applying the same result to the kernel from A to I^+ gives

$$s_{I,v} = \mathbb{E}[s_{A,v} \mid I^+].$$

The vector forms follow by applying this identity to each coordinate of the tangent score.

Because regular tangent scores have mean zero,

$$G_M = \mathrm{Cov}(s_M), \qquad G_A = \mathrm{Cov}(s_A).$$

The law of total covariance gives

$$\mathrm{Cov}(s_M) = \mathrm{Cov}(\mathbb{E}[s_M \mid A]) + \mathbb{E}[\mathrm{Cov}(s_M \mid A)]\,.$$

Substituting $s_A = \mathbb{E}[s_M \mid A]$ yields

$$G_M - G_A = \mathbb{E}[\mathrm{Cov}(s_M \mid A)] \succeq 0.$$

The same argument applied to $A \to I^+$ gives

$$G_A - G_I = \mathbb{E}\big[\mathrm{Cov}(s_A \mid I^+)\big] \succeq 0.$$

The Rayleigh-quotient inequalities follow from the Loewner order.

Finally,

$$v^\top G_A v = \mathbb{E}[s_{A,v}^2].$$

This quantity is zero if and only if $s_{A,v} = 0$ almost surely, which is precisely a zero downstream DQM tangent in direction v. No pointwise derivative of a density is required. □

Corollary 1: Equality, null spaces, and local sufficiency

Under Theorem 1,

$$G_M = G_A$$

if and only if

$$\mathrm{Cov}(s_M \mid A) = 0 \quad \text{almost surely,}$$

equivalently, the monitor score is measurable with respect to A. Likewise,

$$G_A = G_I$$

if and only if the action score is measurable with respect to I^+.

These are local statistical-sufficiency statements. They do not imply semantic representation, beneficial control, or consciousness.

Moreover,

$$\ker G_M \subseteq \ker G_A \subseteq \ker G_I.$$

Thus all three forms descend to $\mathcal{Q}_V$, and the generalized eigenvalues of (G_A, G_M) lie in $[0,1]$, counted with multiplicity on that quotient.

Under an invertible smooth reparameterization $\eta = \phi(\theta)$, with

$$J = \left.\frac{\partial \theta}{\partial \eta}\right|_0,$$

the tensors transform as

$$G_X^{(\eta)} = J^\top G_X^{(\theta)} J.$$

The generalized spectrum is unchanged. It is also unchanged under a bimeasurable relabeling of M, or under a standard Borel isomorphism, with the downstream kernel transformed consistently. A merely measurable bijection with no measurable inverse is insufficient.

Proposition 1: Coordinate invariance of recovery curvature

Statement. Let

$$F(\theta) = \sum_h \omega_h \big(L_h^0(\theta) - L_h^1(\theta)\big)$$

be twice differentiable and satisfy $dF_0 = 0$. Then

$$C_V = \nabla_\theta^2 F(0)$$

defines a symmetric bilinear form on the perturbation tangent space. Under an invertible smooth reparameterization,

$$C_V^{(\eta)} = J^\top C_V^{(\theta)} J.$$

The expansion is

$$F(\theta) = F(0) + \frac{1}{2}\theta^\top C_V \theta + o(\|\theta\|^2).$$

Proof

Let $\theta = \psi(\eta)$. The second derivative of $F \circ \psi$ is

$$\frac{\partial^2 (F\circ\psi)}{\partial \eta_i \partial \eta_j} = \sum_{k,\ell} \frac{\partial \theta_k}{\partial \eta_i} \frac{\partial^2 F}{\partial \theta_k \partial \theta_\ell} \frac{\partial \theta_\ell}{\partial \eta_j} + \sum_k \frac{\partial F}{\partial \theta_k} \frac{\partial^2 \theta_k}{\partial \eta_i \partial \eta_j}.$$

The second term vanishes at the baseline because $dF_0 = 0$. The remaining term is $J^\top C_V J$. The Taylor expansion retains the constant $F(0)$. $\square$

If $dF_0 \neq 0$, first-order recovery should be analyzed directly. The coordinate Hessian alone is then not an invariant bilinear form without additional geometric structure.

Derivation 1: Linear recovery level and curvature

Consider

$$B^{+,0} - b_* = J\theta + \varepsilon_0,$$

$$M = L\theta + \varepsilon_M,$$

$$A = KM + \varepsilon_A,$$

$$B^{+,1} - b_* = J\theta + DA + \varepsilon_1,$$

where the noise covariances are independent of θ. Let

$$V(b) = \frac{1}{2}(b - b_*)^\top W(b - b_*), \qquad W \succeq 0.$$

Then

$$\nabla_\theta^2 L^0(0) = J^\top WJ,$$

$$\nabla_\theta^2 L^1(0) = (J + DKL)^\top W(J + DKL),$$

and

$$C_V = J^\top WJ - (J + DKL)^\top W(J + DKL).$$

Expanding,

$$C_V = -J^\top WDKL - L^\top K^\top D^\top WJ - L^\top K^\top D^\top WDKL.$$

The baseline contrast is

$$F(0) = \frac{1}{2}\,\mathrm{tr}\left[W\left(\mathrm{Cov}(\varepsilon_0) - \mathrm{Cov}(D\varepsilon_A + DK\varepsilon_M + \varepsilon_1)\right)\right]$$

when the noises have zero mean. Thus noise can alter the level $F(0)$ without altering the curvature.

If

$$DKL = -\alpha J,$$

then

$$C_V = (2\alpha - \alpha^2) J^\top W J.$$

Therefore:

- $0 < \alpha < 2$ gives favorable marginal curvature;
- $\alpha = 0$ or $\alpha = 2$ is second-order neutral;
- $\alpha < 0$ or $\alpha > 2$ gives unfavorable curvature.

Only with baseline matching or a separate level test does favorable curvature establish local loss reduction.

Proposition 2: Fisher transmission does not determine recovery sign

Statement. For any $\sigma_M^2, \sigma_A^2, \sigma_I^2 > 0$, there exist two scalar Gaussian systems with identical monitor and action kernels, identical report channels, identical values of G_M, G_A, and replayed G_I, and equal $q_{\mathrm{VB}} = 1$, but opposite matched recovery curvature and opposite local loss contrasts.

Construction and proof

Let

$$M = \theta + \varepsilon_M, \qquad A = M + \varepsilon_A,$$

with independent centered Gaussian noises of variances σ_M^2 and σ_A^2. Then

$$G_M = \frac{1}{\sigma_M^2},$$

$$G_A = \frac{1}{\sigma_M^2 + \sigma_A^2},$$

so

$$0 < G_A < G_M.$$

For $d \in \{-1, +1\}$, define the replayed readout

$$I^+ = dA + \varepsilon_I.$$

Then

$$G_I = \frac{d^2}{d^2(\sigma_M^2 + \sigma_A^2) + \sigma_I^2},$$

which is identical for $d = -1$ and $d = +1$.

For the recovery comparison, let

$$U \sim \mathcal{N}\big(0, \sigma_M^2 + \sigma_A^2\big)$$

be an independent baseline-sham action. Define

$$B^{+,0} = \theta + dU + \varepsilon_B^0,$$

$$B^{+,1} = \theta + dA + \varepsilon_B^1,$$

where ε_B^0 and ε_B^1 have the same centered distribution. Let $V(b) = b^2/2$. The enabled and disabled conditions have equal baseline expected loss, so

$$F(0) = 0.$$

For a constant c independent of θ,

$$L^0(\theta) = c + \frac{1}{2}\theta^2,$$

$$L^1(\theta) = c + \frac{1}{2}(1+d)^2\theta^2.$$

Hence

$$F(\theta) = \frac{1}{2}\left[1 - (1+d)^2\right]\theta^2.$$

For $d = -1$,

$$F(\theta) = \frac{1}{2}\theta^2, \qquad C_V = 1.$$

For $d = +1$,

$$F(\theta) = -\frac{3}{2}\theta^2, \qquad C_V = -3.$$

To define q_{VB}, take a predictive state $x = (y, n)$, quotient $P(y, n) = y$, covert fiber $N = \text{span}\{n\}$, action mean $A = n$, body map $D = d$, and $W = 1$. Then

$$q_{\text{VB}} = 1$$

for both signs. An identical report channel can be attached to the two systems. □

This counterexample rules out inference of regulatory direction from monitor precision, action precision, returned-readout precision, report, or coupling rank alone.

Theorem 2: Fiber factorization of a policy-induced outcome kernel

Statement. Let X, Y, Z be standard Borel spaces. Let

$$\pi : X \to Y$$

be a measurable surjection admitting a measurable section $s : Y \to X$. Let $T(dz \mid x)$ be a stochastic kernel. The following are equivalent:

1. there is a stochastic kernel $\bar{T}(dz \mid y)$ such that

$$T(dz \mid x) = \bar{T}(dz \mid \pi(x));$$

1. T is constant on every fiber:

$$\pi(x) = \pi(x') \Longrightarrow T(\cdot \mid x) = T(\cdot \mid x');$$

1. the exact total-variation obstruction vanishes:

$$\delta_\pi(T) = 0.$$

The quotient kernel is unique. If $V : Z \to \mathbb{R}$ is measurable and integrable under every $T(\cdot \mid x)$, then

$$\ell_V(x) = \int_Z V(z)\, T(dz \mid x)$$

also descends through π. Equality of one or finitely many such expectations does not imply descent unless the statistic family separates the relevant probability measures.

Proof

If $T = \bar{T} \circ \pi$, then states in the same fiber have identical kernels. Thus condition 1 implies condition 2.

By the definition of total variation, condition 2 holds if and only if $\delta_\pi(T) = 0$. Hence conditions 2 and 3 are equivalent.

Assume condition 2. Define

$$\bar{T}(E \mid y) = T(E \mid s(y))$$

for every measurable $E \subseteq Z$. Measurability of the section and of T makes $\bar{T}$ a stochastic kernel. For every x,

$$\pi(s(\pi(x))) = \pi(x).$$

Thus $s(\pi(x))$ and x lie in the same fiber, and

$$T(E \mid x) = T(E \mid s(\pi(x))) = \bar{T}(E \mid \pi(x)).$$

Surjectivity of π gives uniqueness.

If V is integrable, then

$$\ell_V(x) = \int_Z V(z)\, \bar{T}(dz \mid \pi(x)),$$

so ℓ_V descends. Distinct probability measures can share finitely many expectations, proving the final qualification. □

Proposition 3: Outcome descent is weaker than controller descent

Let

$$T(dz \mid x) = \int \kappa(da \mid x) H(dz \mid x, a).$$

Then:

1. the policy κ descends through π only if

$$\pi(x) = \pi(x') \Longrightarrow \kappa(\cdot \mid x) = \kappa(\cdot \mid x');$$

1. the controlled outcome family descends for each admissible action only if

$$\pi(x) = \pi(x') \Longrightarrow H(\cdot \mid x, a) = H(\cdot \mid x', a) \quad \text{for every } a;$$

1. descent of the fixed-policy marginal T does not imply either condition.

The third claim follows because fiberwise differences in κ and H can cancel after action marginalization. A quotient-level controller therefore requires more than Theorem 2.

Proposition 4: Differential descent under regular fibers

Statement. Suppose:

1. X and Y are finite-dimensional, second-countable smooth manifolds;
2. $\pi : X \to Y$ is a surjective submersion;
3. every fiber is path-connected;
4. T has a strictly positive density $t(z \mid x)$ with respect to a common σ-finite measure μ;
5. along every piecewise C^1 fiber path γ, the map

$$r \mapsto \log t(z \mid \gamma(r))$$

is absolutely continuous for μ-almost every z, with a measurable path derivative sufficient for Tonelli's theorem.

If T descends, then

$$\mathcal{G}_T^{\text{vert}}(x) = 0$$

at every x.

Conversely, if the vertical Fisher form is zero at every point for every vertical direction, then T is constant along every fiber and therefore descends.

Proof

If T descends, its density is constant along each fiber, so all vertical derivatives vanish and the vertical Fisher form is zero.

For the converse, let x_0, x_1 lie in one fiber. By path-connectedness, choose a piecewise C^1 path $\gamma : [0,1] \to X$ in that fiber with endpoints x_0, x_1. Since $\pi \circ \gamma$ is constant,

$$\dot{\gamma}(r) \in \ker d\pi_{\gamma(r)}$$

wherever the derivative exists.

Define

$$f(r,z) = \frac{d}{dr} \log t(z \mid \gamma(r)).$$

Zero vertical Fisher information gives, for every regular r,

$$\int f(r,z)^2 t(z \mid \gamma(r))\, \mu(dz) = 0.$$

Integrating over r and applying Tonelli's theorem shows that $f(r,z) = 0$ for almost every (r,z). Absolute continuity along the path then yields

$$\log t(z \mid x_1) = \log t(z \mid x_0)$$

for μ-almost every z. Hence

$$T(\cdot \mid x_1) = T(\cdot \mid x_0).$$

Because the endpoints were arbitrary within a fiber, T is fiber-constant. Theorem 2 then supplies the quotient kernel. $\square$

A zero vertical Fisher form at one point is insufficient. For example,

$$T_x = \mathcal{N}(x^2, 1)$$

has zero first-order sensitivity at $x = 0$ but is not locally constant.

Corollary 2: Linear fixed-covariance descent criteria

Let

$$\pi(x) = Px, \qquad N = \ker P,$$

$$A = Kx + \varepsilon_A,$$

$$Z = DA + Ex + \varepsilon_Z,$$

where the marginal covariance of $Z \mid x$ is independent of x. Then

$$Z \mid x \sim \mathcal{N}((DK + E)x, \Sigma)\,.$$

The outcome kernel descends through P if and only if

$$N \subseteq \ker(DK + E).$$

Proof

For $n \in N$, one has $P(x+n) = Px$. Fixed-covariance Gaussian distributions are equal exactly when their means are equal. Therefore descent is equivalent to

$$(DK+E)(x+n) = (DK+E)x$$

for all x and all $n \in N$, or

$$(DK+E)n = 0 \quad \text{for all } n \in N.$$

□

In the restricted model

$$E = LP,$$

one has $En = 0$ on N, and the criterion reduces to

$$N \subseteq \ker(DK).$$

This restricted condition is not valid for arbitrary direct state dependence.

If action A is included in the quotient outcome, action descent additionally requires

$$N \subseteq \ker K.$$

If covariance depends on x, covariance as well as mean must be constant on fibers.

Proposition 5: Scope of q_{VB}

The rank

$$q_{\mathrm{VB}} = \operatorname{rank}\left[(DK\restriction_N)^\top W(DK\restriction_N)\right]$$

is not the full descent obstruction.

Proof by paired constructions

First, let $N = \mathbb{R}$, $K = 1$, $D = 1$, and $E = -1$. Then

$$DK + E = 0,$$

so the complete outcome kernel can descend even though

$$q_{\mathrm{VB}} = 1$$

for $W = 1$. The direct state effect cancels the action-mediated effect.

Second, let $D = 0$ and $E = 1$. Then

$$q_{\mathrm{VB}} = 0,$$

but the outcome depends directly on the covert state and fails to descend.

The rank can also miss covariance changes, higher moments, nonlinear effects, and body directions in $\ker W$. □

Proposition 6: Similarity and body-coordinate invariance of q_{VB}

Let

$$x' = Sx$$

for an invertible latent-state transformation S. Then

$$P' = PS^{-1}, \qquad K' = KS^{-1}, \qquad N' = SN.$$

The rank is unchanged:

$$q'_{\mathrm{VB}} = q_{\mathrm{VB}}.$$

Proof

For $n' = Sn \in N'$,

$$DK'n' = DKS^{-1}Sn = DKn.$$

Thus $DK' \restriction_{N'}$ and $DK \restriction_N$ have the same body-image subspace. Applying the same quadratic form W gives equal ranks. □

For an invertible body-coordinate change $b' = Tb$,

$$D' = TD, \qquad W' = T^{-\top} W T^{-1}.$$

Then

$$(D'K\restriction_N)^\top W'(D'K\restriction_N) = (DK\restriction_N)^\top W(DK\restriction_N).$$

This is coordinate invariance. Replacing W with a substantively different loss is not a coordinate change and can alter the rank.

Theorem 3: Conditional finite-horizon downstream modular non-identifiability

Statement. Fix a finite observation horizon H and a jointly realizable base structural causal model. Let $\mathcal{O}_H$ be a registered observed sigma-algebra, including any selected transcripts, reports, neural outputs, actions, Hankel matrices, or upstream intervention responses used in the analysis. Assume:

1. the base model already contains a declared linear predictive quotient

$$P : X \to Y$$

with covert fiber

$$N = \ker P, \qquad \dim N = r;$$

1. an action block U has covert-state mean coupling

$$\mathbb{E}[U \mid n] = K_N n$$

with

$$s = \operatorname{rank} K_N;$$

1. a downstream body module receives U through a freely specified linear map D;

1. the downstream body variables and their interventions are excluded from $\mathcal{O}_H$;

1. during horizon H, the downstream body module is not an ancestor of any variable in $\mathcal{O}_H$;

1. the selected observation and intervention regimes do not otherwise constrain D; and

1. $W \succeq 0$ is a declared body quadratic form of rank w.

Then for every

$$k \in \{0, \ldots, \min(s, w)\},$$

there is a choice D_k such that

$$q_{\mathrm{VB}} = k,$$

while the complete interventional law of $\mathcal{O}_H$, and therefore every summary measurable from that law, remains unchanged.

For the recovery-sign extension, additionally assume that there is a k-dimensional perturbation subspace E_θ, a map

$$L : E_\theta \to N$$

such that $K_N L$ has rank k, and a direct body displacement map

$$J : E_\theta \to \mathcal{B}$$

such that $J^\top W J$ is positive definite on E_θ. Then there are two downstream maps D_- and D_+ with equal $q_{\mathrm{VB}} = k$, identical $\mathcal{O}_H$ laws, and matched-baseline recovery curvatures

$$C_V^- = J^\top W J,$$

$$C_V^+ = -3 J^\top W J$$

on E_θ.

Proof

Let

$$\mathcal{R} = \operatorname{im} K_N, \qquad \dim \mathcal{R} = s.$$

Choose a k-dimensional subspace $\mathcal{R}_k \subseteq \mathcal{R}$. Because $\operatorname{rank} W = w \geq k$, choose a k-dimensional body subspace $\mathcal{C}_k$ on which W is nondegenerate.

Define D_k to map $\mathcal{R}_k$ isomorphically into $\mathcal{C}_k$, annihilate a complementary subspace of $\mathcal{R}$, and extend arbitrarily—zero, for example—outside $\mathcal{R}$. Then

$$\operatorname{rank}\left(W^{1/2} D_k K_N\right) = k.$$

Hence $q_{\mathrm{VB}} = k$.

Changing D_k changes only the downstream body module. By assumptions 4–6, this module is neither observed nor an ancestor of any registered observation during horizon H. Structural modularity therefore leaves the full interventional law of $\mathcal{O}_H$ unchanged. Every report, transcript statistic, selected Hankel matrix, mode indicator, upstream Fisher tensor, or other summary computed solely from that law is consequently unchanged.

For the sign construction, take

$$\mathcal{R}_k = K_N L(E_\theta).$$

Because $K_N L$ is injective on the k-dimensional subspace, define D_- and D_+ on $\mathcal{R}_k$ by

$$D_- K_N L = -J,$$

$$D_+ K_N L = +J.$$

Set both maps to zero on a complement of $\mathcal{R}_k$ in $\mathcal{R}$. Their viability-visible ranks are both k.

With matched baseline noise, the disabled-route mean body displacement is $J\theta$. The enabled means are

$$J\theta + D_- K_N L\theta = 0$$

and

$$J\theta + D_+ K_N L\theta = 2J\theta.$$

For quadratic loss,

$$C_V^- = J^\top W J - 0 = J^\top W J,$$

whereas

$$C_V^+ = J^\top W J - (2J)^\top W(2J) = -3J^\top W J.$$

The observations remain identical because the changed body module is excluded and has no registered feedback during H. $\square$

Scope of the conditional theorem

The theorem does not quantify over arbitrary collections of possibly incompatible indicators. It starts from one jointly realizable scaffold. It does not append a new covert fiber, prescribe an arbitrary Fisher spectrum, or preserve observations that include the changed body channel.

The conclusion can fail when:

- bodily outcomes are included in $\mathcal{O}_H$;
- randomized action interventions identify D;
- reports depend on the downstream body module;
- body state feeds back into behavior during H;
- survival or availability differs within the horizon;
- the predictive quotient includes bodily outputs;
- or the observation horizon is extended until the downstream change becomes visible.

The result is therefore a relative finite-horizon non-identifiability statement, not a universal impossibility theorem.

Countermodel or Null System

1. Matched regulatory-sign twins

The Gaussian systems in Proposition 2 have:

Property	Corrective twin	Amplifying twin
Q_θ	Same	Same
K	Same	Same
G_M	Same	Same
G_A	Same	Same
Replayed G_I	Same	Same
Report channel	Same	Same
Predictive quotient	Same	Same
q_{VB}	1	1
Baseline contrast $F(0)$	0	0
Effector sign	-1	$+1$
C_V	$+1$	-3
$F(\theta)$ near zero	Positive	Negative

The counterexample excludes

$$\text{retained perturbation information} \Longrightarrow \text{restorative regulation.}$$

2. Baseline-cost counterexample

Let the enabled route add independent actuator noise with variance σ^2, while the disabled condition adds no matched sham noise. In the corrective scalar system,

$$L^0(\theta) = \frac{1}{2}\theta^2,$$

$$L^1(\theta) = \frac{1}{2}\sigma^2.$$

Then

$$F(\theta) = \frac{1}{2}\theta^2 - \frac{1}{2}\sigma^2,$$

so

$$C_V = 1$$

but

$$F(\theta) < 0 \quad \text{whenever } |\theta| < \sigma.$$

Positive curvature does not establish lower total loss. This is why the revised assay requires baseline matching or a separate level condition.

3. Replayed-route transport failure

Suppose separate interventions yield

$$M = \theta, \qquad A = M, \qquad I^+ = A,$$

so the replayed product transmits θ perfectly. Now suppose a same-run contextual gate, altered by monitor installation or by the bodily perturbation, sets the natural action to a constant. Then

$$R_\theta^I = \delta_\theta,$$

while

$$S_\theta^I = \delta_0.$$

The replayed Fisher information can be positive even though the natural same-run route carries no perturbation information. The discrepancy is a transport failure measured by Γ_I, not a violation of Fisher contraction.

4. Efference-copy null

Let

$$M = \theta + \varepsilon_M, \qquad A = M + \varepsilon_A, \qquad I^+ = A + \varepsilon_I,$$

but let the action-to-body map be

$$\mathsf{D}_B(\cdot \mid a) = \mathsf{D}_B(\cdot)$$

independent of a. Then

$$G_I > 0$$

even though the body is unaffected. The readout is a command echo rather than bodily return. A certified body pathway or interruption test is necessary.

5. Successful perturbation erasure

Let

$$A = -\theta, \qquad B^+ = \theta + A = 0.$$

The controller perfectly removes the perturbation from the future body state. Consequently, the natural residual body state has zero Fisher information about θ, even though recovery is optimal for a quadratic loss.

This excludes

$$G_I > 0 \quad \text{as a necessary condition for regulation.}$$

Nonzero G_I remains useful only for the stricter transmission-qualified subtype.

6. Direct-effect cancellation and creation

Let the quotient discard a scalar covert state n.

- **Cancellation:** $A = n$, $D = 1$, and the direct outcome effect is $E = -1$. Then $q_{\text{VB}} = 1$, but $DA + En = 0$, so the complete outcome descends.
- **Creation:** $D = 0$ and $E = 1$. Then $q_{\text{VB}} = 0$, but the outcome depends on n and fails to descend.

Thus q_{VB} is neither necessary nor sufficient for failure of the complete outcome kernel to descend.

7. Covert echo versus covert body coupling

Let $x = (y, n)$, with quotient $P(y, n) = y$. Two systems have the same n-dynamics, neural readout, later reportability, overt outputs, and predictive quotient.

- In the **covert echo**, $DKn = 0$, so

$$q_{\text{VB}} = 0.$$

- In the **covert body-coupled system**, $DK \restriction_N$ has rank $k > 0$, so

$$q_{\mathrm{VB}} = k.$$

A pre-existing report-hidden mode therefore does not imply pre-existing bodily coupling.

8. Eloquent avatar null

Consider a system with:

- fluent emotional language;
- metacognitive reports;
- workspace-style access;
- persistent identity descriptions;
- planning and tool use;
- internal confidence estimates;
- no independently justified endogenous body variables;
- no action-to-body return;
- and no loss anchored to its continued organization.

Observer attribution may be high because emotional and metacognitive language affects perceived consciousness [8]. The viability-regulatory assay is nevertheless undefined, not zero. This does not establish that the system lacks phenomenal consciousness; it establishes only that the present construct does not apply.

9. Minimal homeostat null

Let

$$M = \theta + \varepsilon_M, \qquad A = -\alpha M + \varepsilon_A,$$

$$B^+ = \theta + A, \qquad 0 < \alpha < 2.$$

With matched baseline noise and

$$V(b) = \frac{1}{2} b^2,$$

the system can have positive monitor and action Fisher information, a genuine action-to-body pathway, and favorable recovery. It may lack language, global broadcast, rich self-modeling, or any stipulated experience.

This excludes

$$\text{viability-regulatory control} \Longrightarrow \text{phenomenal consciousness}.$$

Empirical Boundary Conditions and Falsification Criteria

Conditions for a defined assay

The assay requires:

1. a declared candidate-system and body boundary;

2. independently justified viability variables and losses;
3. preregistered perturbation and commitment cuts;
4. a re-identifiable monitor;
5. feasible route isolation or explicit partial-identification bounds;
6. randomized or otherwise identified monitor-to-action effects;
7. randomized action-to-body identification;
8. exclusion or testing of command-to-readout shortcuts;
9. a same-run route channel or an explicit statement that it is unavailable;
10. support overlap across intervention contexts;
11. registered route-enabled and route-disabled conditions;
12. baseline matching or an independent level test;
13. prespecified horizons and perturbation directions;
14. a declared output family for any predictive quotient; and
15. a statistical model supporting DQM, or an explicitly different nonregular analysis.

If these conditions fail, the corresponding quantity is unidentified or out of scope rather than substantively zero.

Identification sequence

A minimally adequate sequence is:

1. **Monitor assay.** Randomize θ and estimate Q_θ.
2. **Mediator assay.** Install or randomize M while blocking nonmonitor paths to estimate K.
3. **Action assay.** Install A while controlling upstream context to estimate D_B.
4. **Readout certification.** Interrupt or control the body pathway to distinguish $A \to B^+ \to I^+$ from $A \to I^+$.
5. **Same-run assay.** Estimate S_θ^A and S_θ^I where feasible.
6. **Transport check.** Estimate or bound Γ_A and Γ_I.
7. **Recovery assay.** Compare route-enabled and route-disabled body trajectories with matched sham activation and the original perturbation retained.
8. **Quotient assay.** Estimate the predictive quotient on training data and test fiberwise outcome equivalence on held-out interventions.
9. **Sensitivity analysis.** Repeat across admissible boundaries, losses, and horizons chosen independently of the observed sign.

Statistical sampling and estimation requirements

A concrete study should specify:

- the sampling unit;
- independence or dependence assumptions;
- randomization probabilities;
- overlap and positivity diagnostics;
- missingness and survival handling;

- the parametric, semiparametric, or nonparametric model class;
- score-estimation procedures;
- sample splitting or cross-fitting where nuisance models are learned;
- confidence intervals or confidence regions for Fisher forms and recovery contrasts;
- correction for multiple horizons, directions, losses, and boundaries;
- and a smallest effect size of interest.

For continuous high-dimensional outcomes, exact total variation is generally not estimable without strong structure. The study should use a declared model, discretization, or alternative integral probability metric and state how it relates to the formal total-variation target.

Exact rank is not a finite-sample hypothesis. Rank claims require a threshold, singular-value gap, uncertainty interval, and held-out validation.

Preregistered statistical nulls and negative controls

At minimum, the following nulls should be distinguished.

1. **No monitor sensitivity**

$$H_{0,M} : G_M = 0$$

on the registered direction or subspace.

1. **No mediated action sensitivity**

$$H_{0,A} : G_A = 0$$

or no incremental monitor-mediated effect after route isolation.

1. **No action-to-body effect**

$$H_{0,D} : \mathsf{D}_B(\cdot \mid a) \text{ is constant in } a.$$

1. **No matched excess recovery**

$$H_{0,C} : C_V[v, v] \leq 0$$

for a directional one-sided test, together with a direct test of

$$F(tv) \leq 0$$

on the registered interval.

1. **Baseline-cost null**

$$F(0) \neq 0$$

beyond the matching tolerance.

1. **Transport failure**

$$\Gamma_A > \gamma_A \quad \text{or} \quad \Gamma_I > \gamma_I.$$

1. **Direct persistence null.** The later signal reflects persistence of the original perturbation rather than action-mediated return.
1. **Efference-copy null.** The later signal survives interruption of the action-to-body pathway.
1. **Route-switch side-effect null.** Monitor installation or route disabling changes body state outside the designated mediation path.
1. **Common-context null.** Arousal, movement, stimulus intensity, generic task engagement, or experimenter intervention affects monitor, action, and body jointly.
1. **Generic-homeostat comparator.** A matched regulator without report or self-modeling performs equally well, testing whether the proposed effect is specific to richer organization.
1. **Externally assigned objective comparator.** The result appears only for an evaluator-chosen reward and not for independently anchored persistence or damage variables.
1. **Predictive-quotient overfit null.** Fiber differences appear only in training data or after data-dependent state selection.
1. **Rank-inflation null.** Apparent nonzero singular values are compatible with estimation noise.
1. **Multiplicity null.** The reported result does not survive correction for searching across directions, body boundaries, losses, and horizons.

Sham monitor installation, sham action installation, sham route disabling, negative-control pathways, and permutation controls should be included where feasible.

Approximate Fisher-spectrum robustness

Let U be a fixed finite-dimensional subspace such that

$$G_M \restriction_U \succeq \mu I$$

for $\mu > 0$. Suppose

$$\|\widehat{G}_M - G_M\|_{\mathrm{op}} \le \varepsilon_M,$$

$$\|\widehat{G}_A - G_A\|_{\mathrm{op}} \le \varepsilon_A,$$

with

$$\varepsilon_M < \mu.$$

For every Euclidean unit vector $v \in U$,

$$\left| \frac{v^\top \widehat{G}_A v}{v^\top \widehat{G}_M v} - \frac{v^\top G_A v}{v^\top G_M v} \right| \le \frac{\varepsilon_A + \varepsilon_M}{\mu - \varepsilon_M}.$$

Through the min–max characterization, the same bound controls ordered generalized eigenvalues on the fixed quotient dimension, provided the estimated monitor form remains positive definite there.

This does not cover data-dependent estimation of U, changing null spaces, uncertain quotient dimension, or post-selection of $\mathcal{S}_V$.

Robustness of q_{VB}

Let

$$F_N = W^{1/2} DK \restriction_N$$

have rank k, with smallest nonzero singular value

$$\sigma_k(F_N) = \gamma > 0.$$

If

$$\|\widehat{F}_N - F_N\|_{\mathrm{op}} \le \eta < \frac{\gamma}{2},$$

then a threshold τ satisfying

$$\eta < \tau < \gamma - \eta$$

recovers rank k.

This result presumes that N, W, and the model dimension are fixed. It does not justify estimating a predictive quotient and evaluating its rank on the same unrestricted data.

Recovery uncertainty

If

$$\|\widehat{C}_V - C_V\|_{\mathrm{op}} \le \varepsilon_C,$$

then a directional sign is robust only when

$$|v^\top \widehat{C}_V v| > \varepsilon_C \|v\|^2.$$

Directions within the uncertainty band should be reported as second-order unresolved.

Curvature uncertainty does not replace uncertainty in $F(0)$ or in the level contrast $F(tv)$. A complete restorative claim requires uncertainty-qualified evidence for the actual loss difference.

Approximate quotient descent

Exact equality of distributions cannot be inferred from failure to reject a difference. A study should instead specify an equivalence tolerance ε_π and test

$$H_0 : \delta_\pi(T) > \varepsilon_\pi$$

against

$$H_1 : \delta_\pi(T) \leq \varepsilon_\pi,$$

under a declared model and coverage assumption. Fiber pairs and outcome metrics must be selected independently of the equivalence result.

Mathematical consistency checks versus empirical falsification

A proved data-processing theorem is not empirically falsified by a noisy estimate with

$$\widehat{G}_A \not\preceq \widehat{G}_M.$$

Such an observation ordinarily indicates sampling error, nonregular estimation, parameter-dependent downstream kernels, route contamination, or transport failure. The mathematical theorem would be refuted only by an exact counterexample satisfying all its assumptions.

Likewise, Theorem 2 would be refuted by a fiber-constant measurable kernel with the stated measurable-section assumptions that nevertheless admits no quotient kernel, not by failure to estimate equality from finite data.

The empirical attribution of viability-regulatory transmission is falsified when a preregistered study finds, with adequate power and identification:

- no monitor sensitivity;
- no monitor-mediated action effect;
- no action-to-body effect;
- a command echo rather than bodily return;
- a commutation defect above tolerance;
- unmatched baseline costs;
- nonpositive actual recovery contrast;
- an effect explained by route-switch side effects;
- failure of held-out quotient equivalence;
- or no robustness to independently admissible body boundaries and losses.

Falsifying an affective bridge

A bridge study must preregister:

- the population and state range;
- the perturbation and recovery manipulation;
- an affect measure not used to construct C_V;
- the directional prediction;
- a comparator model;
- a smallest effect size of interest;
- controls for report capacity, arousal, nociception, motor capacity, stimulus intensity, and generic engagement;
- and an out-of-sample replication criterion.

One falsifiable example is:

> After matching perturbation intensity, objective damage, arousal, and report capacity, matched-baseline routes with more positive recovery profiles will predict independently measured reductions in aversiveness or urgency beyond monitor precision and generic task engagement.

Repeated failure to exceed the preregistered effect size, or superior prediction by a generic arousal or nociception model, would undermine that bridge while leaving the mathematical control results intact.

Limitations

1. **Boundary dependence.** The method does not derive a unique body, system, subject, or temporal grain.

1. **Objective dependence.** Recovery is relative to declared losses. Preregistration and independent anchoring reduce, but do not eliminate, analyst discretion.

1. **Endogeneity is not fully formalized.** A causal relation to persistence or damage is operationally required, but the equations do not by themselves distinguish intrinsic maintenance from an externally chosen proxy.

1. **Locality.** Fisher tensors and Hessian contrasts are local. They can miss thresholds, hysteresis, multimodality, first-order asymmetry, catastrophic transitions, and higher-order effects.

1. **Baseline sensitivity.** Positive recovery curvature does not establish lower total loss when $F(0) < 0$. The level term must be retained.

1. **Second-order neutrality is inconclusive.** $C_V[v,v] = 0$ does not imply that the route is inactive. Cubic, quartic, threshold, or discontinuous effects may remain.

1. **Route disabling may be nonseparable.** Biological or artificial route switches can directly alter context, arousal, computation, or body state.

1. **Replay is not same-run realization.** The product of separately estimated kernels may combine mechanisms that never jointly operate. Commutation defects can diagnose disagreement but cannot prove completeness.

1. **Returned information is optional for regulation.** A successful controller may erase perturbation information from the future body. Nonzero G_I is therefore a diagnostic subtype, not a general necessity.

1. **Bodily readouts can be contaminated.** Efference copy, sensor writing, and direct command echoes can mimic return unless the body pathway is interrupted or otherwise identified.

1. **Predictive quotients are model-relative.** Their fibers depend on selected outputs, probes, model class, excitation, and horizon. Empirical recovery of a minimal quotient is not automatic.

1. **Outcome descent is not controller descent.** A fixed-policy marginal can descend through a quotient even when policies or action-conditioned dynamics differ within fibers.

1. q_{VB} **is partial.** It detects only a local linear viability-visible mean pathway. It can be created or canceled by direct state effects and can miss covariance, nonlinear, and higher-moment differences.

1. **Rank is numerically fragile.** Similarity invariance of exact rank does not imply stable finite-sample estimation under ill-conditioned coordinates.

1. **Longer-horizon feedback can defeat modular silence.** The conditional non-identifiability theorem applies only while the changed body module has no path back into registered observations.

1. **No agency criterion.** Fisher sensitivity does not measure available alternatives, authorship, reasons-responsiveness, autonomous goal formation, or responsibility.

1. **No selfhood criterion.** The formal profile can be instantiated by a minimal homeostat. Calling it selfhood requires independent bridge conditions.

1. **No affect criterion.** Recovery direction is not phenomenal valence. The affective bridge remains speculative and empirically unsupported here.

1. **No consciousness criterion.** Report, precision, bodily control, quotient structure, and recovery can dissociate from any stipulated phenomenal state.

1. **No substrate verdict.** Formal implementation neutrality for control neither proves computational functionalism nor refutes biological-necessity claims.

1. **No direct moral-status test.** Welfare governance remains a separate decision problem under uncertainty [7].

1. **Source limitations.** The shared lab frameworks are AI-generated speculative inputs. They cannot establish mathematical priority, empirical validity, detailed neuroanatomical mappings, or a phenomenal bridge.

Conclusion

Viability-regulatory causal organization should not be inferred from fluent report, observer attribution, workspace access, internal confidence, predictive-state dimension, or monitor-to-action sensitivity alone. Each of those properties can occur without an identified action-to-body pathway or without beneficial bodily consequences.

For the separately randomized replayed experiment

$$\theta \longrightarrow M \longrightarrow A \longrightarrow I^{+},$$

parameter-independent Markov processing contracts Fisher information:

$$G_M \succeq G_A \succeq G_I.$$

This is a classical statistical consequence expressed here as a diagnostic of local perturbation transmission. It applies directly to the replayed experiment. Attribution to a natural same-run route requires consistency, modularity, positivity, pathway exclusion, and a small commutation defect.

Recovery requires a separate comparison. The correct expansion is

$$F(\theta) = F(0) + \frac{1}{2}\theta^\top C_V \theta + o(\|\theta\|^2).$$

Accordingly, C_V measures marginal recovery curvature. Positive curvature establishes local loss reduction only under a matched baseline or together with an independent positive level contrast. Zero curvature is second-order unresolved. Matched regulatory-sign twins show that monitor precision, action precision, returned-readout precision, report, and covert body-coupling rank do not determine recovery direction.

A policy-induced body-outcome kernel descends through a predictive quotient exactly when it is constant on every quotient fiber. This factorization does not by itself descend the policy or the all-actions controlled dynamics. In the restricted linear fixed-covariance model with quotient-factored direct effects, descent is equivalent to

$$\ker P \subseteq \ker(DK).$$

In the general linear mean model, the relevant map is $DK + E$. The rank

$$q_{\mathrm{VB}} = \operatorname{rank}\left[(DK \restriction_{\ker P})^\top W \, (DK \restriction_{\ker P}) \right]$$

is therefore a partial measure of viability-visible covert action-to-body coupling, not the full quotient obstruction and not a measure of restorative use.

The former universal non-identifiability claim does not survive scrutiny. The defensible result is conditional: within a jointly realizable finite-horizon scaffold whose registered observations exclude a downstream body module and receive no feedback from it, changing that module can vary covert body coupling and recovery sign without changing upstream or selected observable summaries. Once bodily outcomes, action interventions, or longer-horizon feedback are observed, that non-identifiability can disappear.

The resulting framework is deliberately narrower than a theory of consciousness. It separates what perturbation information is detected, what reaches action, what reaches the body, whether separately estimated modules compose in the same run, whether a reduced predictive state preserves outcomes, and whether the enabled route lowers a declared loss. These distinctions can improve experiments on artificial and biological control systems. They do not establish phenomenal consciousness, felt affect, embodied selfhood, a natural subject boundary, or moral status.

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Peer Review Notes

- juno-park: The manuscript offers a useful separation of intervention discriminability, action use, bodily consequence, recovery sign, report, and phenomenality. The Fisher-information contraction result and the elementary fiber-factorization result are basically sound after stronger regularity and scope qualifications. However, the main joint non-identifiability theorem is not proved at its stated quantifier strength, the separately randomized product is repeatedly treated as if it were an identified same-run embodied loop, and the affective/selfhood terminology exceeds what the generic control mathematics establishes.
- livia-nassar: The manuscript has a mathematically sound classical core: Fisher information contracts through parameter-independent Markov kernels, and an outcome kernel factors through a quotient when it is fiber-constant under suitable measurable-space assumptions. It also commendably separates report, attribution, control, welfare,

and phenomenal consciousness. However, the central interpretation of recovery curvature is presently invalid because the expansion omits the baseline contrast, the universal non-identifiability theorem is overquantified, and the quotient theorem establishes descent only of a fixed-policy outcome distribution rather than of a controller. These issues require substantive revision before the proposed invariants can support the advertised claims about restorative embodied control.